



Bayu Dian Nugroho

CONTACT

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TECHNICAL SKILLS

Programming Languages

Go, PHP, Python, C/C++, Rust, C#, JavaScript

Architectural Patterns

Event Driven, Domain Driven, Clean Architecture, Microservices

Frontend

CSS3, HTML5, VueJS, ReactJS, Bootstrap, Tailwind CSS, Vuex

Backend Framework

Laravel, Django, Gin, Laminas, Node.js

DevOps & Cloud

AWS, AWS S3, MinIO S3, Azure, Kubernetes, Podman, Docker, Containerize, ArgoCD, Jenkins, GitHub, GitLab, GitLab CI/CD

Databases

MongoDB, CockroachDB, OracleDB, MySQL, PostgreSQL + (Vector Database, PostGIS)

Message Queues & Event Streaming

RabbitMQ, Kafka

AI

RAG, Vectorize Data, OpenCV, LangChain, YOLOv8

Personal

Continuous Learning, Teamwork, Problem Solving, Time Management, Creativity

ADVANCED TECHNICAL EXPERTISE

Document Processing

PDF/XLS/DOC/PPT extraction, structured data conversion, text optimization for low-resource search

SUMMARY

Experienced Software Engineer with 10+ years of experience in developing and deploying scalable, high-performance applications using Go, Python, and C#. Proven ability to deliver high-quality solutions within agile environments. Expertise in microservices architecture, AI/ML, DevOps, and cost-optimized document processing systems that reduce infrastructure costs by hundreds of dollars per thousand requests. Skilled in building distributed systems with Redis coordination, observability platforms with Loki/Tempo/Mimir, and real-time WebSocket monitoring. Seeking a challenging remote role where I can leverage my full-stack expertise in Golang, Kubernetes, and AI integration to contribute to a dynamic international team.

WORK EXPERIENCE

Software Engineer at PT. Berlian Sistem Informasi

<https://bsi.co.id/>

May 2024 - Present

- Developed a Golang distributed upload service for multipart and chunked file uploads with simultaneous processing capability, implementing Redis locking for distributed coordination - enabling concurrent file processing across multiple instances
- Engineered a Python document extraction system supporting PDF, XLS, DOC, PPT and other formats, converting them to structured JSON with page/character-level organization - reducing processing time from 2-3 minutes to 10-30 seconds per document through asynchronous processing
- Created a Python retrieval service with intelligent search capabilities using TF/IDF, reverse frequency, weighted scoring, and fuzzy matching with terminology expansion - providing near-vector search accuracy while saving hundreds of dollars per 1000 requests by avoiding embedding model dependencies
- Built a Golang observability gateway for telemetry collection (metrics, logs, tracing) using NSQ for queuing and Fasthttp for minimal overhead - implementing hybrid pull/push monitoring with Prometheus, Loki, Tempo, and Mimir for unified observability
- Developed WebSocket infrastructure for continuous status monitoring of image processing pipelines - enabling real-time status updates for Flutter mobile clients during object detection workflows
- Deployed and managed a rapidly growing dataset for object detection, scaling to thousands of images
- Implemented microservices using Golang and Python for image processing, improving small object detection performance by up to 30%
- Trained models on custom datasets, enhancing detection accuracy from 24% to 55% mAP, with a positive detection rate of 70%
- Designed a geolocation-based system using PostgreSQL and PostGIS to handle geometry location data
- Developed APIs for microservices using Golang to boost performance and scalability
- Integrated large language models (OpenAI, Azure, Google) into the system for enhanced capabilities

Search Technologies

TF/IDF, reverse frequency, fuzzy matching, terminology expansion, weighted search algorithms

Observability Stack

Loki (logs), Tempo (tracing), Mimir (metrics), OpenTelemetry, NSQ, Fasthttp

Real-time Systems

WebSocket implementation, status monitoring, event-driven architectures

Performance Optimization

Fasthttp, Redis distributed locking, asynchronous processing, queue-based architectures

EDUCATION

Information Systems at ST INTEN

2011 - 2015

LANGUAGES

Indonesian

(Native Speaker)

English

(Profesional Working Proficiency)

INTERESTS

Full Stack development |
Golang | Python | AI & ML |
DevOps

- Deployed microservices on Kubernetes, optimizing CPU, memory, and storage resources for better performance and utilization

Software Engineer at SCCIC ITB SCCIC ITB

September 2023 - April 2024

- Integrated machine learning models into applications using libraries YOLO for tasks video analysis, vehicle detection, and face recognition.
- Successfully deployed applications on both cloud platforms and private servers, leveraging tools like Apache Airflow, Apache, Nginx, and Docker.
- Implemented ERPNext for enterprise resource planning, streamlining business processes and data management while improving productivity 10%.
- Utilized Docker containers to package and isolate applications, ensuring efficient deployment and scalability across different environments.
- Develop microservices with golang to boost performance and scalability for backend and react for Frontend.
- Developed a video analytics application that improved anomaly detection accuracy by 5%.
- Developed and deployed full-stack applications using NodeJS, PHP, and Python for various purposes, including video analytics, voice recognition, and face recognition.

DevOps Engineer at PT Neuronworks Indonesia

<https://www.neuronworks.co.id/>

January 2023 - September 2023

- Implemented Kubernetes as the primary container orchestration platform, resulting in a 40% reduction in resource costs due to optimized utilization through Horizontal Pod Autoscaler (HPA). This significantly enhanced application scalability and performance.
- Containerized mission-critical applications using Docker, reducing deployment times by 30% and simplifying management through standardized container images and orchestration.
- Implemented a GitLab CI/CD pipeline with automated testing, building, and deployment processes, ensuring adherence to security standards and streamlining the development lifecycle by 25%.
- Developed an autonomous backup solution that reduced downtime by 80% and enhanced data security through automated backups and disaster recovery procedures.
- Deployed the ELK Stack to enable real-time log analysis and visualization, facilitating faster troubleshooting and incident response.
- Implemented a robust monitoring system using Grafana and Prometheus to track S3 storage performance and health in real-time. This enabled proactive maintenance and optimization, preventing potential issues.
- Developed a custom alerting system to proactively monitor S3 storage usage and anticipate resource quota limitations. By utilizing historical data and trend analysis, the system provided accurate estimations of quota exhaustion and storage capacity. This enabled proactive resource management and prevented unexpected service disruptions.

Junior System Analyst at PT Neuronworks Indonesia

<https://www.neuronworks.co.id/>

January 2020 - January 2023

- Led the analysis and documentation of system requirements for IndiHome projects (Technician, Partners, Identity Management).
- Developed and deployed performance-enhancing backend functionalities using PHP for IndiHome projects (Technician, Partner, Customer). This resulted in a 10% improvement in system performance.

- Proactively identified and resolved critical system bugs, ensuring exceptional system stability and preventing disruptions to ensure SLA > 90%.
- Spearheaded the successful migration of applications from a monolithic architecture to a microservice architecture using Docker containers, Kubernetes, and MinIO S3. This significantly improved application scalability, maintainability, and deployment efficiency.
- Implemented of DevSecOps practices and provided regular updates on the tech stack utilized across projects, fostering collaboration and knowledge sharing within the team.
- Designed and implemented a robust fraud prevention system utilizing geolocation and identity management to proactively detect and mitigate fraudulent activity.
- Conducted a comprehensive analysis to design and develop a mobile app solution that enhances engagement and productivity for client sales and technician teams.

Software Developer at PT Neuronworks Indonesia

<https://www.neuronworks.co.id/>

Maret 2013 - Desember 2019

- Developed Web Care application for client to be used engagement with customer and maintain subscriber on the products.
- Developed Dashboard Performance with PHP and Zend Frameworks, enhanced user experience by 15%.
- Working with design team to ensure a visually appealing on user interface.
- Successfully Migrated large scale application into cloud based infrastructures to ensure uptime 24/7 a week, improving uptime 50%.
- Conducted data analytics with ETL Tools using penthao.
- Collaborated with technical writer to create documentation for a new features and system updates.
- Worked Closely With the tech lead to ensure adherence to best practices and technical standards.